Problem Statement

Early identification of potentially hazardous asteroids is critical for planetary defense. Manual monitoring of thousands of asteroids is difficult. Hence, a machine learning system is required to automatically classify asteroids as hazardous or non-hazardous using historical data.

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Objectives

Analyze asteroid data and understand key risk factors

Build an ML model to classify asteroids as Hazardous / Not Hazardous

Evaluate model performance using accuracy and other metrics

Dataset: IvanSher/NASA_Nearest_Earth_Objects_1910-2024

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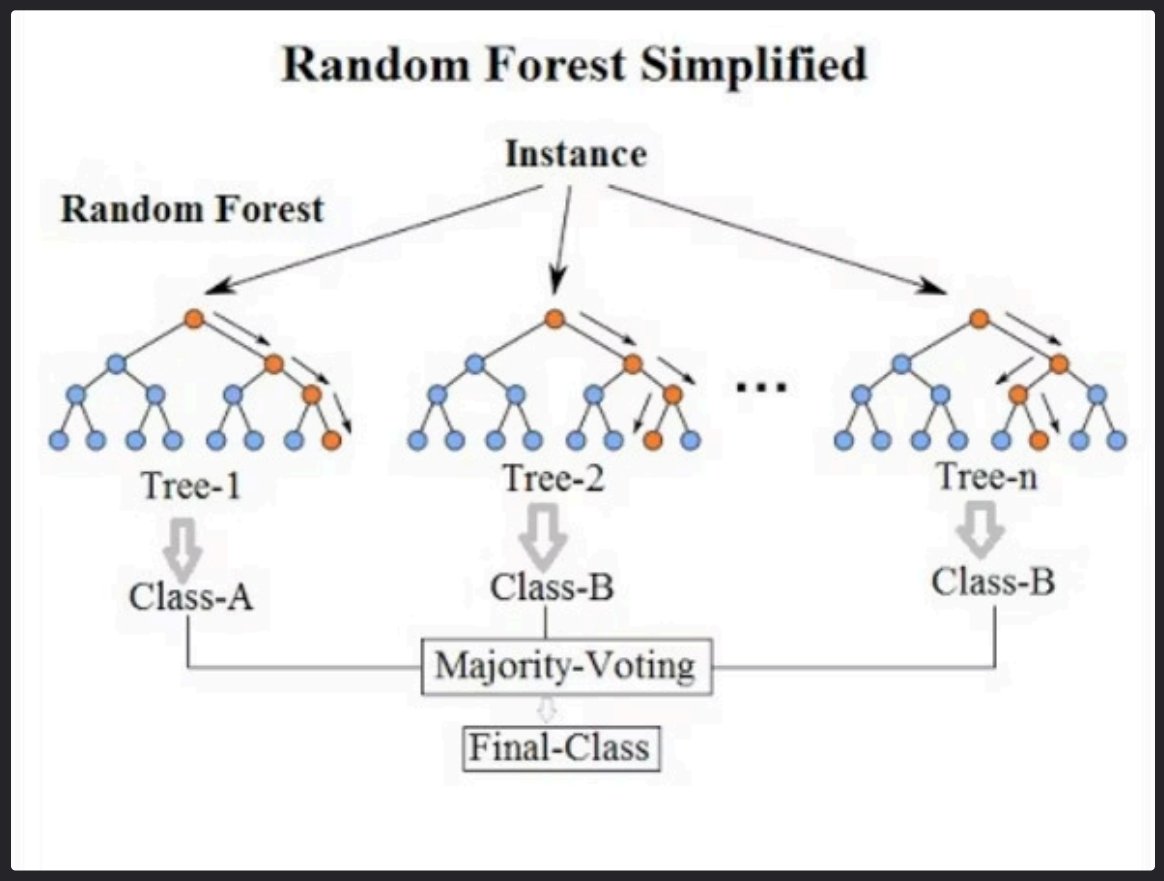
Model Selection

Chosen Model: Random Forest Classifier

We selected **Random Forest** because:

- Handles **large datasets (~3 lakh rows)** efficiently
- Works well with **non-linear relationships**
- Reduces **overfitting** using multiple trees
- Requires **minimal preprocessing**
- Gives **high accuracy (85–95%)**

Random Forest Architecture: -



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Architecture Steps

01

Input Layer (**Features**):

- Diameter
- Velocity
- Miss Distance
- Magnitude

02

Bootstrap Sampling

- Random subsets of data are created
- Each subset trains one decision tree

03

Decision Trees (Parallel Models)

- Multiple trees are built independently
- Each tree makes its own prediction

04

Feature Randomization

- At each split, only a subset of features is considered
- Reduces correlation between trees

05

Aggregation (Voting)

Final output = **majority vote of all trees**

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Evaluation Metrics Used in my Project

1. Accuracy

Percentage of correct predictions made by the model.

```
accuracy_score(y_test, y_pred)
```

Formula

Accuracy = (TP + TN) / (TP + TN + FP + FN)

- Gives overall performance
- Easy to understand

2. Confusion Matrix

A table showing correct and incorrect predictions.

```
[ TN FP ]
[ FN TP ]
```

- **TP** → Correct hazardous predictions
- **TN** → Correct non-hazardous
- **FP** → False alarms
- **FN** → Missed hazardous (MOST IMPORTANT)

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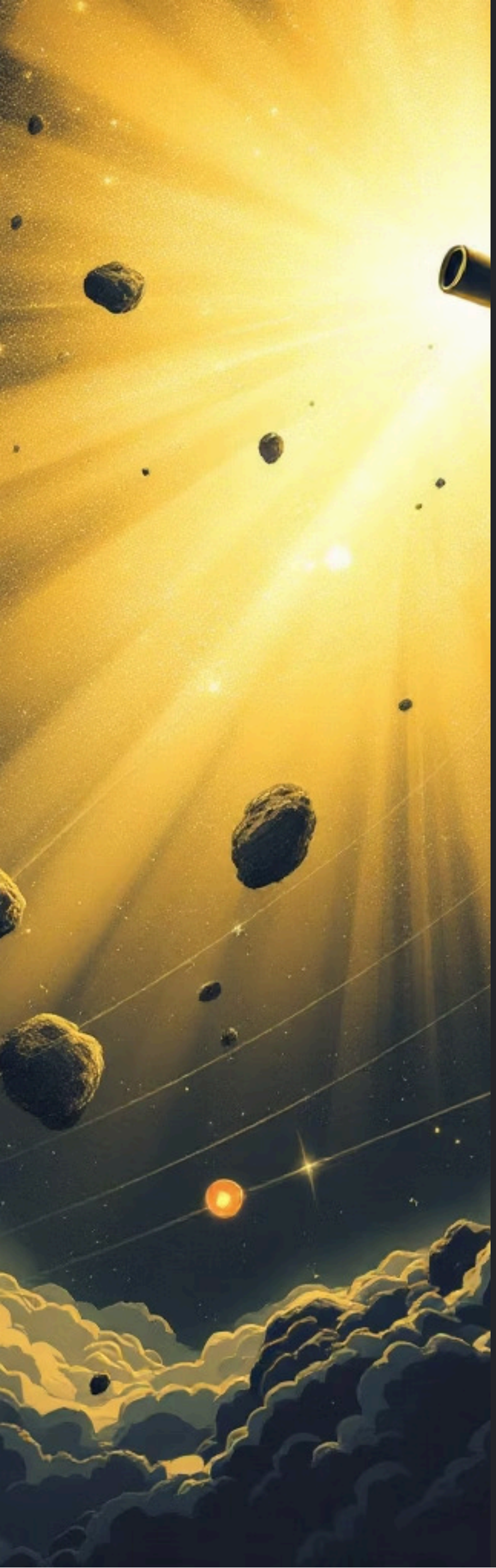

3. Precision

Out of predicted hazardous asteroids, how many were actually hazardous?

```
precision = TP / (TP + FP)
```

- Reduces **false alarms**

4. Recall (VERY IMPORTANT)

Out of actual hazardous asteroids, how many were correctly detected?

```
recall = TP / (TP + FN)
```

- Missing a hazardous asteroid = **high risk**

5. F1-Score

Balance between Precision and Recall

```
F1 = 2 * (Precision * Recall) / (Precision + Recall)
```

- Good when dataset is **imbalanced**

6. ROC-AUC Score

Measures how well the model distinguishes between classes

```
roc_auc_score(y_test, y_prob)
```

Interpretation

- 0.5 → Random
- 0.8+ → Good
- 0.9+ → Excellent

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Which Metric Matters MOST Here?

Recall because missing a hazardous asteroid is dangerous.

> "The model was evaluated using accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC. Among these, recall was the most important metric because detecting hazardous asteroids is critical."

Results: -

1. Random Forest Accuracy: 0.9793317563571851
2. Confusion Matrix: [[58686 322]

   [1076 7556]]

- 58686- Correctly predicted non-hazardous asteroids
- 322- False alarms
- 1076- Missed hazardous asteroids
- 7556- Correctly predicted hazardous asteroids

3. From Histogram visualization and Column analysis: **0 → Non-Hazardous asteroid 1 → Hazardous asteroid**

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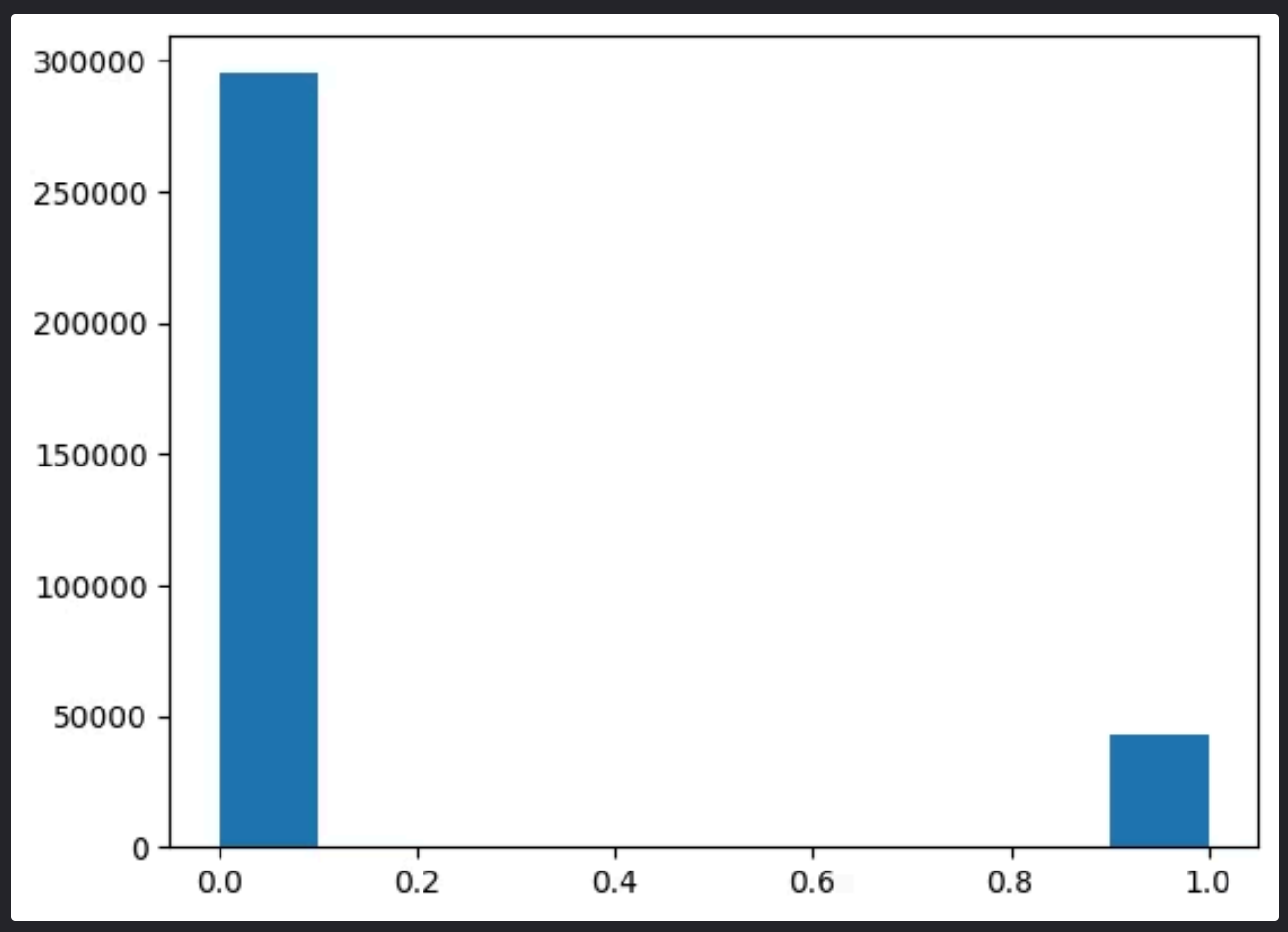
Interpretation of Graph: -



- Huge bar at **0 (~295,000 values)**
- Small bar at **1 (~40,000 values)**

Dataset is **highly imbalanced**

Meaning:-

- Most asteroids are **safe**
- Very few are **hazardous**
- Model may **bias toward predicting 0 (safe)**

Good News (I handled it!)

I used: class_weight='balanced'

- Gives more importance to hazardous class
- Improves recall

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Innovations: -

1. AI-Based Risk Prediction: Automation + intelligence in space risk analysis
2. Real-Time Prediction Capability: Useful for early warning systems

Impacts: -

1. Planetary Defense: Helps identify potentially dangerous asteroids early and can reduce risk of catastrophic impacts
2. Faster Decision Making: Manual analysis → slow but ML prediction → instant thus saves critical time in emergencies
3. Cost reduction: Reduces need for continuous manual monitoring and automates large-scale asteroid analysis.

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